NEUROVOX — Voice-Based Parkinson's Screening

Java Programming Course End Project · Exam Notes

1. Aim of the Project

To build a Java app that listens to a person's voice, measures how **shaky** it is (pitch, jitter, shimmer, HNR) and tells whether the person is at **low risk** or **high risk** of Parkinson's disease — a quick, cheap and painless early check.

2. Description of the Project

- **The problem:** Parkinson's is a brain disease that slowly damages the cells that control movement. It is usually found **very late** — when shaking starts, most of those cells are already gone.
- **Why it is hard to catch early:** Checking for it needs a specialist doctor, many tests and costly scans. These are slow and **not easy to get in villages**. There is no simple early test.
- **What we noticed:** Almost **9 out of 10 patients get voice changes**, often early. Their voice becomes shaky, weak and breathy.
- **Our idea:** Use a normal microphone as a **free check-up tool**. Just saying "aaah" for 5 seconds is enough to see how steady the voice is.
- **How we solved it:** We measure 4 things — **Pitch** (how high/low), **Jitter** (pitch shaking), **Shimmer** (loudness shaking) and **HNR** (how clean vs noisy the voice is). More shaking + more noise = more risk.
- **Cleaning the audio:** Our Java server (JDK HttpServer) makes the volume even, cuts out silent parts and breaks the sound into small pieces (frames).
- **Finding the pitch:** We wrote the **autocorrelation** method ourselves in Java — it slides the sound over itself to find where it repeats. We compared it with another method called **YIN**.
- **Checking our work:** We matched our Java results with **Praat**, a standard voice-analysis tool, to make sure our numbers are right.
- **The prediction:** A **logistic regression** model trained on the **UCI Parkinson's dataset** (195 voice recordings from 31 people). We tested it on **new people only**, so the accuracy is honest and not fake.
- **Result:** Saved in an **SQLite** database and shown on a web dashboard (HTML/CSS/JS). It only says **low or high risk** — it does not replace a doctor.

Key Formulas (write these for extra marks)

Pitch (F0)	`F0 = Sample Rate / Pitch Period`
Jitter (%)	`( avg |T(i) - T(i-1)| ) / avg T  x 100` `(T = period)`
Shimmer (%)	`( avg |A(i) - A(i-1)| ) / avg A  x 100` `(A = amplitude)`
Prediction	`z = b + w1*jitter + w2*shimmer + w3*HNR ;` `P = 1 / (1 + e^(-z))`

3. Main Program 1 — Main.java (Main File: runs the screening)

Entry point of the project. Calls FeatureExtractor to get the features, then uses **logistic regression** (weighted sum → **sigmoid**) to get a risk probability between 0 and 1. Probability ≥ 0.5 → High Risk.

```
public class Main {

    // Weights learned by training on the UCI Parkinson's dataset
    static double bias = 4.0;
    static double wJitter = 2.0;
    static double wShimmer = 0.5;
    static double wHnr = -0.3;

    // Logistic regression: probability = 1 / (1 + e^-z)
    static double predict(double jitter, double shimmer, double hnr) {
        double z = bias + wJitter * jitter + wShimmer * shimmer + wHnr * hnr;
        return 1.0 / (1.0 + Math.exp(-z));
    }

    static String classify(double prob) {
        if (prob >= 0.5) return "HIGH RISK - consult a neurologist";
        else return "LOW RISK";
    }

    public static void main(String[] args) {
        // Values measured from the user's voice recording
        double[] periods = {300, 304, 298, 302, 306};     // in samples
        double[] amps = {0.80, 0.76, 0.82, 0.78, 0.81};   // peak heights
        double hnr = 21.5;                                 // in dB

        // Step 1: Feature extraction
        double f0 = FeatureExtractor.pitch(44100, 302);
        double jitter = FeatureExtractor.variation(periods);
        double shimmer = FeatureExtractor.variation(amps);

        // Step 2: Prediction
        double prob = predict(jitter, shimmer, hnr);

        // Step 3: Result
        System.out.println("---- NEUROVOX SCREENING ----");
        System.out.printf("F0      : %.1f Hz%n", f0);
        System.out.printf("Jitter  : %.2f %%%n", jitter);
        System.out.printf("Shimmer : %.2f %%%n", shimmer);
        System.out.printf("Risk    : %.2f -> %s%n", prob, classify(prob));
    }
}
```

Output

```
---- NEUROVOX SCREENING ----
F0      : 146.0 Hz
Jitter  : 1.49 %
Shimmer : 5.35 %
Risk    : 0.96 -> HIGH RISK - consult a neurologist
```

4. Main Program 2 — FeatureExtractor.java (Important File: Pitch, Jitter, Shimmer)

Finds the pitch period by **autocorrelation** (compare the signal with a shifted copy of itself — the shift with the best match is one period). One **variation()** method gives both jitter and shimmer.

```
public class FeatureExtractor {

    // Find pitch period (in samples) using autocorrelation
    public static int findPeriod(double[] frame, int minLag, int maxLag) {
        int bestLag = minLag;
        double bestCorr = 0;
        for (int lag = minLag; lag <= maxLag; lag++) {
            double sum = 0;
            for (int i = 0; i + lag < frame.length; i++) {
                sum += frame[i] * frame[i + lag];
            }
            if (sum > bestCorr) {
                bestCorr = sum;
                bestLag = lag;
            }
        }
        return bestLag;
    }

    // Fundamental frequency F0 = sampleRate / period
    public static double pitch(int sampleRate, int period) {
        return (double) sampleRate / period;
    }

    // Variation % = avg change between neighbours / average value * 100
    // Used for Jitter (on periods) and Shimmer (on amplitudes)
    public static double variation(double[] x) {
        double diff = 0, total = 0;
        for (int i = 1; i < x.length; i++) {
            diff += Math.abs(x[i] - x[i - 1]);
        }
        for (double v : x) {
            total += v;
        }
        double avgDiff = diff / (x.length - 1);
        double avg = total / x.length;
        return (avgDiff / avg) * 100;
    }
}
```

Conclusion

Neurovox shows that simple Java (arrays, loops, classes, Math) can turn a 5-second voice clip into a useful health check. It is free, fast and needs only a microphone, so people can see a doctor early. **Future scope:** more voice features, bigger datasets, a mobile app.